THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A model is a: 1

(a) unit **(b)** final truth
(c) complete copy **(d)** simplified representation

2. A mixture has components in: 1

(a) one ratio only **(b)** any ratio **(c)** a fixed ratio **(d)** no ratio

3. Mass is conserved in a: 1

(a) nuclear change only **(b)** chemical reaction
(c) only physical change **(d)** no reaction

4. The electron has a: 1

(a) positive charge **(b)** double charge **(c)** no charge **(d)** negative charge

5. A theory is: 1

(a) well-tested explanation **(b)** a symbol
(c) a guess **(d)** a unit

6. Separation of sand from water uses: 1

(a) filtration **(b)** sublimation **(c)** distillation **(d)** chromatography

7. A positive ion is called: 1

(a) anion **(b)** cation **(c)** atom **(d)** molecule

8. The K shell holds at most: 1

(a) 2 electrons **(b)** 8 electrons **(c)** 18 electrons **(d)** 32 electrons

9. 2000 m equals: 1

(a) 2 km **(b)** 200 km **(c)** 20 km **(d)** 0.2 km

10. The mass of solution is: 1

(a) zero **(b)** mass of solute + mass of solvent
(c) mass of solvent only **(d)** mass of solute only

11. The bond in NaCl is: 1

(a) hydrogen **(b)** ionic **(c)** metallic only **(d)** covalent

12. Isotopes have the same: 1

(a) mass number **(b)** atomic number
(c) number of nucleons **(d)** number of neutrons

13. SI unit of length: 1

(a) gram **(b)** litre **(c)** second **(d)** metre

14. Air is a: 1

(a) element **(b)** mixture of gases **(c)** compound **(d)** pure substance

15. The law of conservation of mass was given by: 1

(a) Lavoisier **(b)** Dalton **(c)** Newton **(d)** Proust

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is a model? 2

17. What is a mixture? 2

18. State the law of conservation of mass. 2

19. Name the sub-atomic particles with their charges. 2

20. What is an estimate? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Write three SI base units and quantities. 3

22. List three differences between a solution and a suspension. 3

23. Write three postulates of Dalton's atomic theory. 3

24. State three observations of Rutherford's experiment. 3

OR

List three examples of chemical changes. 3

25. Find the percentage: 20 g salt in 180 g water. 3

OR

List three properties of covalent compounds. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the scientific method with an example. 5

OR

Classify mixtures with two examples each. 5

27. State and explain the law of conservation of mass with an example. 5

OR

Describe Thomson's model of the atom. 5

END OF QUESTION PAPER — Total: 50 Marks

MINDPOINT